

JYOTSNA VENKATESAN

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Education

The Hong Kong Polytechnic University

BSc (Hons) in Computer Science

2022 – 2026

Full Tuition Scholarship

Honors and Awards

Full Tuition Scholarship — Awarded for academic merit (HKD 580,000)	2022 – 2026
HKSAR Reaching Out Award — For international student outreach (HKD 10,000)	2024
Cathay Pacific Hackathon Semi-finalist — Semi-finalist among 1500+ participants	2024
Dean's Honours List — For exceptional academic achievement	2024 – 2025
Micro Fund Shortlisted Team — Admitted to HKSTP Ideation Program and granted HKD 20K	2025
PolyU Delegation, IAC Sydney — Selected to represent The Hong Kong Polytechnic University at International Astronautical Congress	2025

Skills

Programming Languages: Python, Java, C++, C, JavaScript, TypeScript, Swift
Web and Frontend: React, Vue, Svelte, HTML/CSS, Tailwind CSS, Figma
Backend and Databases: Node.js, MySQL, MongoDB, REST APIs
DevOps and Tools: Docker, Git, Bash, MIPS
AI/ML: LLMs, Model Context Protocol (MCP) servers, Azure OpenAI, Claude, GPT APIs
Testing: Selenium, Appium, JUnit, TestNG

Experience

Co-founder and CTO

May 2025 – Present

Orbityr

Hong Kong

- Founded an AI-powered location intelligence company, overseeing product development and technical strategy.
- Accepted into HKSTP's 1-year Ideation Programme (HKD 100,000 grant) and secured additional HKD 20,000 funding.

Full Stack Gen AI Software Developer

June 2025 – August 2025

Career Hackers

Hong Kong

- Developed features for an AI interview platform with TypeScript using the MERN stack.
- Designed data schemas and interfaces for the platform with Figma.
- Built a resume parser, a Model Context Protocol server, and implemented search techniques including BM25, HNSW, and RRF using ElasticSearch.

Quality Assurance Automation Intern

September 2024 – December 2024

GoGoX

Hong Kong

- Automated 90% of regression tests, reducing manual testing time from 2.5 days to 3 hours using Java and Appium.
- Improved test coverage through data-driven testing techniques.

Quality Assurance Intern

June 2024 – August 2024

GoGoX

Hong Kong

- Automated 100% of sanity test suite, reducing execution time by 25% with Java and Appium.
- Expanded automated test coverage across 3 platforms and 3 countries.
- Participated in scrum meetings, gained understanding of agile methodologies.

UI/UX Designer

February 2024 – May 2024

PolyU Smart Cities Research Institute

Hong Kong

- Designed QC Spatial-3Dx, a 3D camera application, using Figma and Adobe Photoshop.
- Led frontend development using React and Three.js for a responsive website.

Web/Frontend Design Engineer Intern

June 2023 – August 2023

Oxbridge Economics

Hong Kong

- Designed and developed web apps using Figma and React.
- Implemented RESTful API connections, improving data flow efficiency between frontend and backend.

Student Assistant

February 2023 – April 2023

CLP

Hong Kong

- Assisted in development and testing of PolyU's Generator Inspection Robot at CLP's power plant.

Frontend Web Development Intern

June 2021 – July 2021

Sleep Deprived Dreamers

India

- Designed and developed the website's frontend using Figma and Knorish (education platform builder).
- Awarded a scholarship of INR 15,000 (approximately HKD 1,300) for contributions.

Projects

Orbityr - Location Intelligence

SvelteKit, TypeScript, Node.js, Tailwind CSS, MCP

- Built an AI-powered location intelligence platform enabling real-time spatial data analysis using SvelteKit and TypeScript for seamless frontend/backend integration.
- Implemented Model Context Protocol based infrastructure for efficient real-time data retrieval and AI model orchestration.
- Designed scalable full-stack architecture with Tailwind CSS for responsive, modern UI.

Secure Online Storage System

Python, AES-GCM, SQLite3, PBKDF2, TOTP, Flask

- Developed a client-server secure file storage system with end-to-end AES-GCM 256-bit encryption, two-factor authentication, and strict access controls.
- Implemented audit logging, parameterized queries, and robust defenses against SQL injection, path traversal, and unauthorized access.

Sporkify - Recipe Management Application

Vue.js, Node.js, SQLite3, Tailwind CSS, Axios, Express.js

- Developed a full-stack recipe management app with secure user authentication, CRUD operations, and responsive UI.
- Implemented advanced search, category-based filtering, and image uploads with RESTful API endpoints using Node.js and Express.
- Designed structured SQLite3 database with tables for users, recipes, and metadata for efficient querying.

Cosmic Soundscape Generator

Python, MIDI, Computer Vision, NumPy, OpenCV, music21

- Generated MIDI music from images using computer vision techniques to map color, texture, and shapes to tempo, scale, instruments, and chord progressions.
- Created multi-track compositions with algorithmic harmony; output project **referenced by Samsung Research India** for integration and citation.

ZZZAFARI - Sleep Tracker Mobile App

SwiftUI, Swift, Core Data, Xcode, Krea

- Developed a sleep tracker mobile app with an interactive animal reward system dynamically reflecting user sleep patterns.
- Created all custom animal assets and AI-assisted backgrounds using Krea to enhance UI/UX design.
- Implemented persistent storage with Core Data and interactive SwiftUI animations for engaging user experience.

Leadership & Activities

Member, Rewriting the Code (Women in Tech community)

2025 – Present

President, PolyU Hall Community Service Group

2024 – 2025

Vice President, PolyU SAO International Student Association

2023 – 2024

Powertrain Team Member, PolyU E-Formula Racing

2023 – 2024

Technical Mapping Team Member, PolyU Robocon

2022 – 2023

Workshop Facilitator, ProjeKt Inspire (Tanzania)

2024